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Effects of vanadium content on the high temperature oxidation behavior of NbTiZrAlV refractory complex concentrated alloys

Yi Lin[a,b,*], Yi Guo[a], Quan Dong[c], Rui Huang[d], Jun Tan[c,**]

[a] School of Materials Science and Engineering, Sichuan University of Science & Engineering, Zigong 643000, China
[b] Material Corrosion and Protection Key Laboratory of Sichuan province, Zigong 643000, China
[c] School of Materials Science and Engineering, Chongqing University, Chongqing 400044, China
[d] School of Materials Science and Engineering, Dalian University of Technology, Dalian 116024, China

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ABSTRACT

Oxidation behavior and microstructural evolution of NbTiZrAlV refractory complex concentrated alloys (RCCAs) with vanadium (V) concentration of 3 and 11 at% were investigated at 500 and 800 °C in the air atmosphere. When NbTiZrAlV RCCAs were exposed at 500 °C, the oxidation behavior was controlled by internal oxygen (O) diffusion, and single oxide layer predominating by Ti_2ZrO_6 and $Nb_2Zr_6O_{17}$ were produced after oxidation for 200 h. The mass gains of alloys with 3 and 11 at% V were 0.042 and 0.018 mg/mm^2, and corresponding average thickness of oxide layer was 70 and 30 μm. As oxidation temperature was increased to 800 °C, volatilization of V_2O_5 brought out the aggravation of oxidation degree for NbTiZrAlV RCCA, and complex oxide layers consisting of Nb_9VO_{25}, $Nb_2Zr_6O_{17}$ and TiO_2 were obtained after thermal exposure for 24 h. The mass gains of alloys with the V content of 3 and 11 at% were 1.135 and 0.514 mg/mm^2, and the average thickness of oxide layer was 500 and 300 μm, respectively. In both cases, the antioxidant performance of NbTiZrAlV RCCA was significantly improved by the increased content of V. And the enhanced oxidation resistance could be ascribed to the fact that the increase of V addition enhanced the atom diffusion resistance via minimizing mixing enthalpy and maximizing configurational entropy.

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1. Introduction

One design strategy with multi–principal metal elements in near–equiatomic concentration was creatively proposed by Cantor's [1] and Yeh's [2] research groups in 2004. Since then, complex concentrated alloys (CCAs) have gradually entered the scene and exhibits unexpected comprehensive properties, such as ultrahigh hardness [3], superior wear resistance [4], excellent corrosion resistance [5], and extraordinary strength and ductility [6]. Refractory complex concentrated alloy (RCCA) has been regarded as one promising candidate to replace traditional high–temperature alloy and received extensive attention since it was firstly reported by Senkov et al. in 2010 [7]. The chemical composition of RCCA is predominated by four or more refractory metals (e.g., Ti, V, Cr, Zr, Nb, Mo, Hf, Ta, and W), and sometimes with other additional elements such as C, Ni, Al or Co [8–10]. Therefore, RCCA usually possesses remarkable high–temperature mechanical properties, and a series of NbTaTiV, NbMoTaWVCr, HfNbTaTiZrMoW, and VNbMoTaW RCCAs present much higher yield strength at the temperature above 1000 °C than Haynes 230 and Inconel 718 do [11–15]. Up to now, numerous outstanding works were done to pursue desirable mechanical properties for RCCAs. For instance, Liaw et al. quantitatively analyzed the distorted lattice of NbTaTiV RCCA, and found that local severe lattice distortion led to the slow elemental diffusion in this alloy at elevated temperatures, which brought out the preferable resistance toward high temperature softening [16]. Tunes et al. fabricated one high hardness (22.8 GPa) $Nb_{23}Ta_{23}Mo_{29}W_{25}$ thin film with enhanced mechanical damage tolerance by ion beam sputter–deposition [11]. Long et al. pointed out that the synthetic action of solid solution hardening of matrix, fine grain strengthening, and interstitial solid solution hardening from O endowed extraordinary combination of yield strength (3416 MPa) and ductility (5.3%) to NbMoTaWVCr alloy at room temperature [12]. Wei et al. declared that the natural–mixing characteristics among refractory elements minimized the segregation of as–cast $Ti_{38}V_{15}Nb_{23}Hf_{24}$ alloy, leading to admirable ductility and physicochemical stability [17]. Cao et al. reported that

* Corresponding author at: School of Materials Science and Engineering, Sichuan University of Science & Engineering, Zigong 643000, China.
** Corresponding author at: School of Materials Science and Engineering, Chongqing University, Chongqing 400044, China
E-mail addresses: llinyi@163.com (Y. Lin), jun.tan@cqu.edu.cn (J. Tan).

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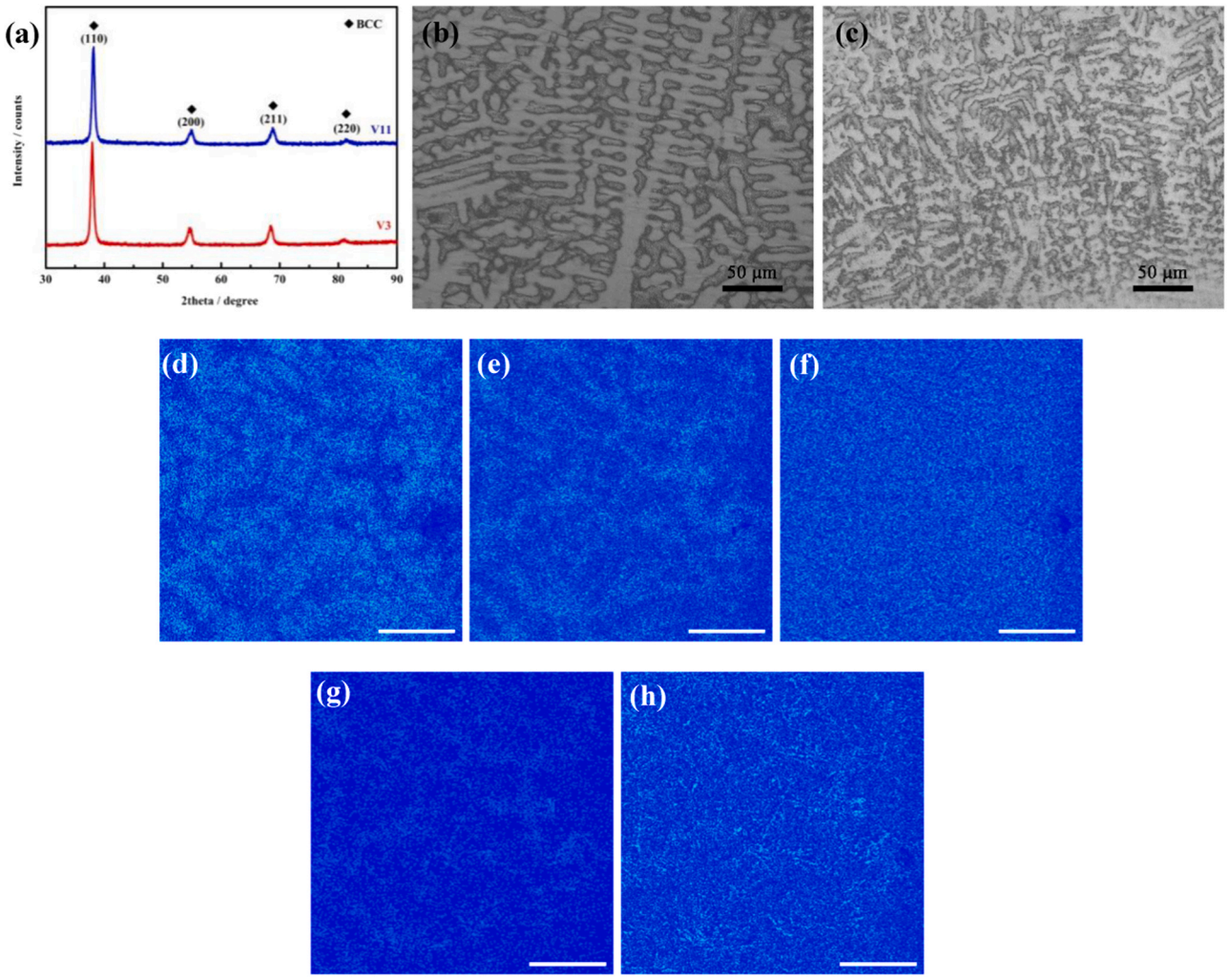


Fig. 1. (a) XRD patterns of the as–cast alloys; OM images of V3 (b) and V11 (c) alloys; Nb (d), Zr (e), Ti (f), Al (g), and V (h) distribution in V11 alloy. The scale bar of (d–h) is 50 μm.

the competition between solid solution hardening and precipitation strengthening dominated the mechanical property of TiNb-$Ta_{0.5}ZrAl_{0.5}$ alloy at room temperature, whereas the reducing in solid solution hardening and grain boundary sliding gave rise to the degradation of yield strength at elevated temperature [18].

Antioxidant performance of RCCA, which ensures the advantage of mechanical properties can be still exerted in high temperature environment, is critical to realize the future utilization in the fields of energy generation, petrochemical industry, and aerospace due to the increasingly hash performance requirement of material. For example, nuclear fuel cladding must be insensitive to high temperature (1200 °C) water vapor oxidation to ensure the nuclear reactor safety [19]; steam turbine is able to withstand the erosion of steam over 700 °C [20]; storage tank of molten salt for solar power generation should be tough in the working temperature range of 300–600 °C [21,22]; high pressure turbine blade in aircraft ought to have high threshold value for oxidation at above 1100 °C [23]. Nevertheless, more researchers pay much more attention to the mechanical properties at elevated temperatures rather than the high temperature oxidation performance of RCCA nowadays; and the investigations probing into the key factor of dominating the oxidation behavior is scarce, especially the mechanism behind how the critical factor influences the antioxidant performance. Additionally, the oxidation behavior may vary greatly among different RCCA systems ascribing from the diversity in chemical composition [24–26]. Thus, exhaustive studies, which focus on the essence of metallic element determining the oxidation behavior of RCCA, have important practical significance toward accurate design of chemical composition and high–efficiency regulation of antioxidant performance for RCCA. According to the prior research, element V, which can provide a large solute misfit volume in body–centered cubic (BCC) and face–centered cubic (FCC) RCCAs, leads admirable strengthening effect to these two crystalline structures. Such as, Dong et al. presented that the yield strength of $AlCoCrFeNiV_x$ alloy reached the peak value of 1726.5 MPa as V molar ratio increased to 0.2 [27]. Wang et al. found that the yield strength and ductility of VNbMoTa alloy at room temperature were greatly enhanced by the increase of V concentration due to solid solution strengthening and grain refinement [28]. Yin et al. established the mathematical model for quantitative predicting the relationship between V content and yield strength, and declared that the optimal strength of CoCrFeMnNiV and MoNbTaVW alloys could be obtained when the atomic concentration of V was around 25% [29]. However, few studies discuss the influence of V on the oxidation behavior of NbTiZrAlV RCCA nowadays to our knowledge. That is to say, the intrinsic relationship between V addition and antioxidant ability for NbTiZrAlV RCCA is far from being fully understood. Herein, NbTiZrAlV RCCAs with the variation of V concentration were fabricated, and microstructural evolution and oxidation behavior during thermal exposure were observed by different analysis technics. Finally, the function of V addition on the oxidation mechanism was discussed in detail, which is expected to shed new

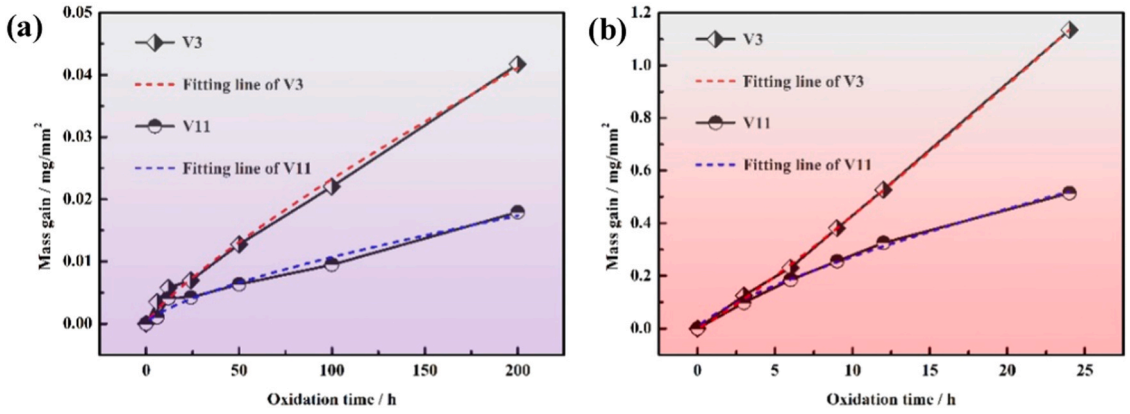


Fig. 2. Oxidation kinetics curves of alloys at 500 (a) and 800 °C (b).

light on the exploration of oxidation mechanism and precise regulation of antioxidant performance for RCCAs.

2. Experimental

Two RCCAs with nominal compositions (in at%) of $Nb_{31}Ti_{34}Zr_{26}Al_6V_3$ (denoted as V3) and $Nb_{31}Ti_{26}Zr_{26}Al_6V_{11}$ (marked as V11) were fabricated using high purity metals (99.99 wt%) in non–consumable vacuum arc furnace. Each prepared alloy was remelted at least seven times to obtain chemical component homogenization.

For isothermal oxidation testing, specimen in the dimension of 10 × 5 × 5 mm was polished with SiC abrasive paper and ultrasonically cleaned in ethanol solution. Isothermal oxidation testing was conducted at 500 and 800 °C in air atmosphere by using Nabertherm LM412 furnace. The sample was individually mounted in corundum boat, placed at the central of furnace during oxidation, and then took out for mass measurement by Mettler Toledo XP6 electronic scale with an accuracy of 0.1 μg at a given time.

The crystal structure was measured by X–ray diffractometer (Buker D2 PHASER) equipped with Cu Kα radiation. The X–ray diffraction (XRD) pattern was recorded in the 2θ range of 20–90° with a scan rate of 0.1° S^{-1}. Metallographs of RCCA before and after oxidation were obtained using optical microscope (OM, Zeiss Axio Vert A1). Electron probe microanalysis (EPMA) was carried out using a JEOL JXA-8230 instrument to obtain a precise elemental distribution in as–cast sample. The cross–sectional morphology and element distribution of oxidized specimen were examined via scanning electron microscope (SEM, JEOL JSM-7500F) equipped energy dispersive X–ray spectrometry (EDS). The dominating oxides were observed by transmission electron microscope (TEM, JEOL JEM-2100F).

3. Results and discussion

3.1. Microstructure of the as–cast NbTiZrAlV RCCAs

XRD patterns of the as–cast alloys are shown in Fig. 1a. Both V3 and V11 alloys exhibited simple BCC solid solution structure, which was usually observed in TiAlZrNb [30], MoNbTaVW [31], HfNbTaTiZr [32], and AlCrMoNbTiZr [33] RCCAs. Meanwhile, the right–shift of XRD curve was detected as the V content was increased from 3 to 11 at%. According to Bragg's law, this change could be ascribed to the fact that more V with smaller atomic radius participated in constructing the BCC RCCA. OM images of the as–cast alloys are displayed in Fig. 1b and c, where V3 and V11 alloys present dendritic structures and the secondary dendrite arm spacing of the former was distinctly larger than that of the latter, suggesting the increase of V addition enhanced the degree of supercooling for the growth of dendritic tip during solidification. According to the EPMA results of V11 alloy (Fig. 1d and e), the dendritic region was enriched in Nb; and interdendritic region was filled with Zr, Al and V; whereas, Ti was homogenous in the alloy.

3.2. Oxidation behavior of NbTiZrAlV RCCAs

Fig. 2 shows the mass gain curves of V3 and V11 alloys during isothermal oxidation at 500 and 800 °C. After oxidized at 500 °C for 200 h, weight gains of V3 and V11 alloys were 0.042 and 0.018 mg/mm^2. The oxidation degree of both alloys became more severe as the temperature increased to 800 °C. And the corresponding mass gains of V3 and V11 alloys after thermal exposure for 24 h were 1.135 and 0.514 mg/mm^2. To uncover the oxidation kinetics of alloys at different oxidation conditions, the measured data were fitted by formula $(\Delta M)^n = kt$ [34], where ΔM stands for mass change per unit area, n is oxidation rate exponent, k describes oxide growth rate, and t represents exposure time. The fitting lines and results are listed in Fig. 2 and Table 1. At 500 °C, the oxidation behaviors of both alloys based on parabolic law, indicating the dense oxide films were obtained on the surface of alloys. At 800 °C, mass gain curves of two alloys were still in accordance with parabolic law, the curve for V3 alloy opened upwards along with the oxidation time, whereas the curve for V11 alloy exhibited the opposite trend, suggesting that oxide film of V3 alloy could not support sufficient protection to the matrix, whereas antioxidant performance of V11 alloy was superior to that of V3 alloy at this condition. Meanwhile, the k values at 500 and 800 °C were reduced by 75% and 63% when the atomic content of V increased from 3% to 11%, demonstrating that antioxidation properties of NbTiZrAlV RCCA was enhanced by the increase of V content.

Fig. 3 shows the XRD patterns of V3 and V11 alloys after isothermal oxidation. It is found that NbTiZrVAl RCCAs exhibited different oxidation behavior at different temperature. At 500 °C, an oxide layer dominated by Ti_2ZrO_6 (JCPDS No: 46-1265) and $Nb_2Zr_6O_{17}$ (JCPDS No: 09-0251) was obtained on two alloys after 200 h (Fig. 3a). When the samples were exposed at 800 °C, oxide composite layer consisted of Nb_9VO_{25} (JCPDS No: 49-0289), $Nb_2Zr_6O_{17}$, and TiO_2 (JCPDS No: 83-2242) could finally form on these

Table 1
The oxide growth rate k and oxidation rate exponent n of alloys at different conditions.

Alloy	500 °C@200 h		800 °C@24 h	
	k (mg^n/mm^{2n} h)	n	k (mg^n/mm^{2n} h)	n
V3	0.0016	1.2136	0.0467	0.8985
V11	0.0004	1.4347	0.0174	1.3532

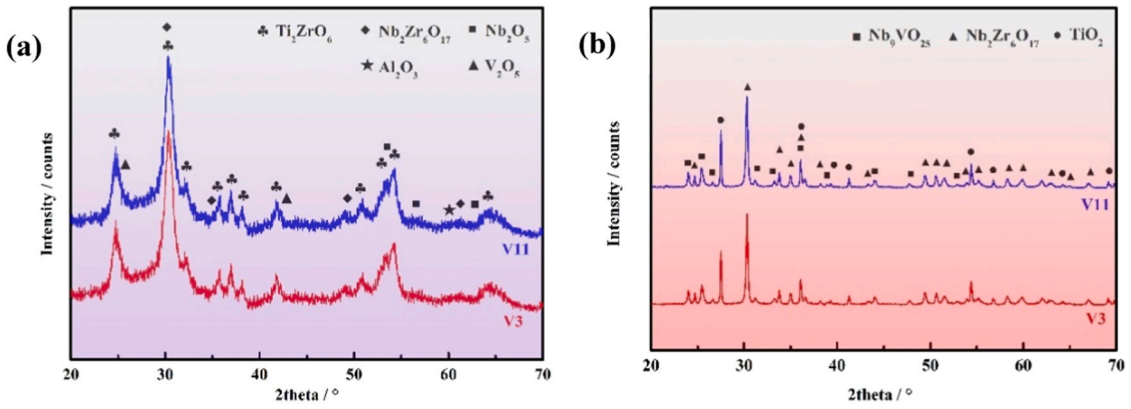


Fig. 3. XRD patterns of NbTiZrAlV RCCAs after isothermal oxidation at 500 °C for 200 h (a) and 800 °C for 24 h (b).

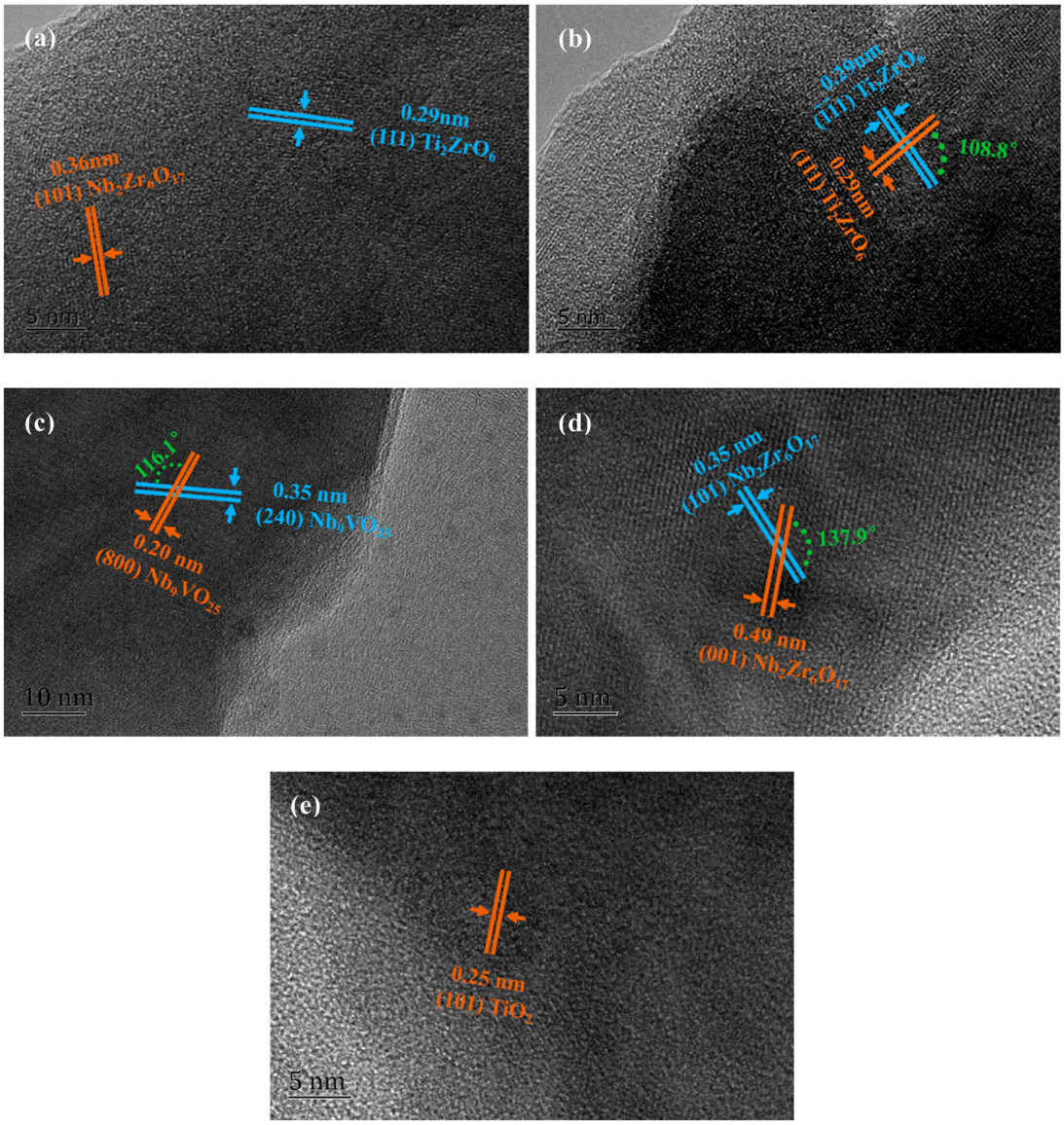


Fig. 4. HRTEM images for dominating oxides obtained at 500 °C for 200 h (a and b) and 800 °C for 24 h (c–e).

two alloys after 24 h (Fig. 3b). Meanwhile, the peak strength and half–peak breadth of XRD patterns were rather low and broad for the sample oxidized at 500 °C, which reflects the low crystallinity of oxide layer, suggesting that the NbTiZrAlV RCCAs possessed excellent oxidation resistance at 500 °C. As the temperature was raised to 800 °C, the oxidation resistance of alloys was extremely deteriorated, so that sharp peak and narrow half–peak breadth were observed in the corresponding XRD patterns. TEM analysis was also

carried out to observe the oxide film, and HRTEM images for dominating oxides formed at different oxidation conditions on V11 alloy were presented in Fig. 4. After oxidation at 500 °C for 200 h, HRTEM image exhibited two kinds of lattice fringes with d-spacing of 0.36 nm and 0.29 nm, which could be assigned to the (101) $Nb_2Zr_6O_{11}$ and (111) Ti_2ZrO_6 (Fig. 4a and b). As exposure at 800 °C for 24 h, the lattice fringes with interplanar distances of 0.20 and 0.35 nm and an intersection angle of 116.1° could be well indexed to the (800) and (240) planes of Nb_9VO_{25} (Fig. 4c). Meanwhile, the lattice fringes with d-distances of 0.49 and 0.35 nm and an intersection angle of 137.9° could be assigned to the (001) and (101) planes of $Nb_2Zr_6O_{11}$ (Fig. 4d). Furthermore, lattice fringes of (101) TiO_2 was also observed (Fig. 4e). These results well confirmed the formation of major oxides on the NbTiZrAlV RCCA alloy during the oxidation process.

To better understand the formation process of oxides at different temperatures, the standard Gibbs free energy of formation for Nb_2O_5, V_2O_5, ZrO_2, TiO_2 and Al_2O_3 at 500–800 °C is given as follow [35]:

$$Ti + O_2 = TiO_2,\ \Delta G = -941 + 0.18T \quad (1)$$

$$4V + 5O_2 = 2V_2O_5,\ \Delta G = -579 + 0.13T \quad (2)$$

$$Zr + O_2 = ZrO_2,\ \Delta G = -1092 + 0.18T \quad (3)$$

$$2Nb + 5O_2 = Nb_2O_5,\ \Delta G = -775 + 0.17T \quad (4)$$

$$4Al + 3O_2 = 2Al_2O_3,\ \Delta G = -1122 + 0.22T \quad (5)$$

where, ΔG represents standard molar Gibbs free energy of oxidation reaction at temperature *T*. The corresponding ΔG values were calculated and listed in Table 2. The descending order of ΔG for oxides was $V_2O_5 > Nb_2O_5 > TiO_2 > Al_2O_3 > ZrO_2$. That is, the stability of oxides increased in the order of $V_2O_5 < Nb_2O_5 < TiO_2 < Al_2O_3 < ZrO_2$. Hence, the oxide formation procedure at different temperature could be described as follow. At 500 °C, ZrO_2 and TiO_2 (rutile) with high thermodynamic stability were initially formed on the surface of sample. Owing to similar unit cell (octahedron) of ZrO_2 [3] and TiO_2 [36] and approximate radius of Zr^{4+} (0.72×10^{-10} m) and Ti^{4+} (0.68×10^{-10} m) [37], ternary oxide Ti_2ZrO_6 was speculated to compound through the reaction between TiO_2 and ZrO_2 in the mole ratio of 2–1:

$$2TiO_2 + ZrO_2 = Ti_2ZrO_6 \quad (6)$$

Meanwhile, ZrO_2 would react with Nb_2O_5 to produce $Nb_2Zr_6O_{17}$ during a long time oxidation reaction, which was also found in the thermal exposure of NbTiZrCr RCCA [36,38]. For element Al, its content in the alloy was rather low. Meanwhile, the pre–exponential factor of Al diffusion (2.5×10^{-4} m^2 s^{-1}) as well as the diffusion activation energy of Al (366 kJ mol^{-1}) in Nb alloy were larger than those of Zr (3×10^{-6} m^2 s^{-1}, 301 kJ mol^{-1}) and Ti (2.5×10^{-7} m^2 s^{-1}, 263 kJ mol^{-1}) [39]. Basing on the Arrhenius formula [40], the diffusion coefficient of Al in Nb alloy was estimated to be 3.7×10^{-29} m^2 s^{-1}, which was approximately two and five order magnitude lower than those of Zr (1.1×10^{-27} m^2 s^{-1}) and Ti (5.0×10^{-24} m^2 s^{-1}). Similarly, the diffusion coefficient of element V in the Nb alloy could be calculated to be 4.4×10^{-28} m^2 s^{-1} with pre–exponential factor of 1.6×10^{-6} m^2 s^{-1} and diffusion activation energy of 410 kJ mol^{-1} [39]. Thus, only a small amount of Al_2O_3 and V_2O_5 were produced in the oxide layer after thermal exposure at 500 °C for 200 h (The diffusion coefficients of Al, Zr, Ti, V in Nb alloy were summarized in Table 3).

Table 2
Standard Gibbs free energy of oxide formation at 500 and 800 °C.

Temperature/°C	Standard Gibbs free energy (ΔG)/kJ mol^{-1}				
	TiO_2	V_2O_5	ZrO_2	Nb_2O_5	Al_2O_3
500	-801.9	-478.5	-952.9	-625.5	-951.9
800	-747.9	-440.9	-894.9	-575.2	-891.4

Table 3
The summary of diffusion coefficients of Al, Zr, Ti, V in Nb alloy.

Diffusion coefficient/m^2 s^{-1}	Al	Zr	Ti	V
	3.7×10^{-29}	1.1×10^{-27}	5.0×10^{-24}	4.4×10^{-28}

When the sample was exposed to air at 800 °C, metallic elements are more likely to combine with O to form simple oxides (e.g., TiO_2, V_2O_5, ZrO_2 and Nb_2O_5) within a short reaction time. Thus, $Nb_2Zr_6O_{17}$ and Nb_9VO_{25} could be synthesized via the solid–state reaction of Nb_2O_5 with ZrO_2 and V_2O_5, respectively [36,38]. In the meantime, equilibrium concentration of lattice vacancy, which is beneficial to the migration of atom in the sample, was sharply increased relative to that at 500 °C. Hence, Al^{3+} and Nb^{5+} could overcome the diffusion barrier to partly substitute the lattice Ti^{4+} in TiO_2. However, the radius of Al^{3+} (0.53×10^{-10} m) was much less than that of Ti^{4+}, simultaneously, the crystal structure of Nb and Al were quite different from that of TiO_2. As a result, small amount of $Ti_{0.6}Nb_{0.2}Al_{0.2}O_2$ might form in the oxide layer after oxidation at 800 °C for 24 h.

Fig. 5 shows the surface morphology of NbTiZrAlV RCCAs after oxidation at different conditions. Clearly, serious peeling–off phenomenon occurred in the surface oxide of V3 alloy after thermal exposure at 500 °C for 200 h (Fig. 5a and b). With the increase of V content, the oxide layer of V11 alloy exhibited a planar surface with grinding mark, and few ruptures were found in the oxide layer (Fig. 5c and d). The surface oxide exhibited different morphology characteristic when the sample was oxidized at 800 °C (Fig. 5e–h). Under this condition, the resultant oxide film consisted of numerous micro columnar crystals, and obvious cracking was observed everywhere in the oxide layer of V3 and V11 alloys. However, the crack density of former with fine columnar crystal was dramatically denser than that of latter. In general, the growth stress and thermal stress bring about the spallation of oxide layer in the oxidation procedure that further promotes the diffusion of O atom to the fresh metallic material, resulting in the aggravation of oxidation degree for material. Accordingly, it can be concluded that the resistance of oxide layer to the growth stress and thermal stress was enhanced along with the increasing amount of V, and which was conductive to improve the antioxidant ability of NbTiZrAlV RCCA at elevated temperature.

The corresponding EDS testing was carried out to analyze the surface chemical composition of oxide layers obtained at different oxidation conditions, and the result is listed at Table 4. After oxidation at 500 °C for 200 h, the atomic ratios among metallic elements in the oxide layers of both alloys were comparable to those in the matrix of two alloys, which demonstrates that the oxidation procedure of NbTiZrAlV alloy at 500 °C was dominated by the O internal diffusion. In addition, the atomic ratio of Ti to Zr in the oxide layer was nearly 1, which reflects that TiO_2 formed at this condition could be completely consumed by the solid–state reaction with ZrO_2 to produce Ti_2ZrO_6. Besides, the residual ZrO_2 in the scale was depleted via the chemical reaction with Nb_2O_5 to form $Nb_2Zr_6O_{17}$. Hence, ZrO_2 and TiO_2 could not be found in the oxide layer after heated treat at 500 °C. When the sample was exposed at 800 °C, the atomic ratios among metallic elements in the oxide layer deviated from the normal design value of NbTiZrAlV alloy, and the degree of deviation became more apparent with the increase of V, especially the concentration of Nb and Zr. This characteristic was an indicator for the outward diffusion of metallic elements in NbTiZrAlV alloy during the oxidation procedure at 800 °C. Meanwhile, binary oxides, such as TiO_2, V_2O_5, ZrO_2 and Nb_2O_5, quickly form on the surface at this condition. Thereinto, the volatilization of V_2O_5 was inevitably took place because of low melting point (ca. 690 °C) [41]. In this case, the elevated partial pressure of V_2O_5 accelerated the reaction

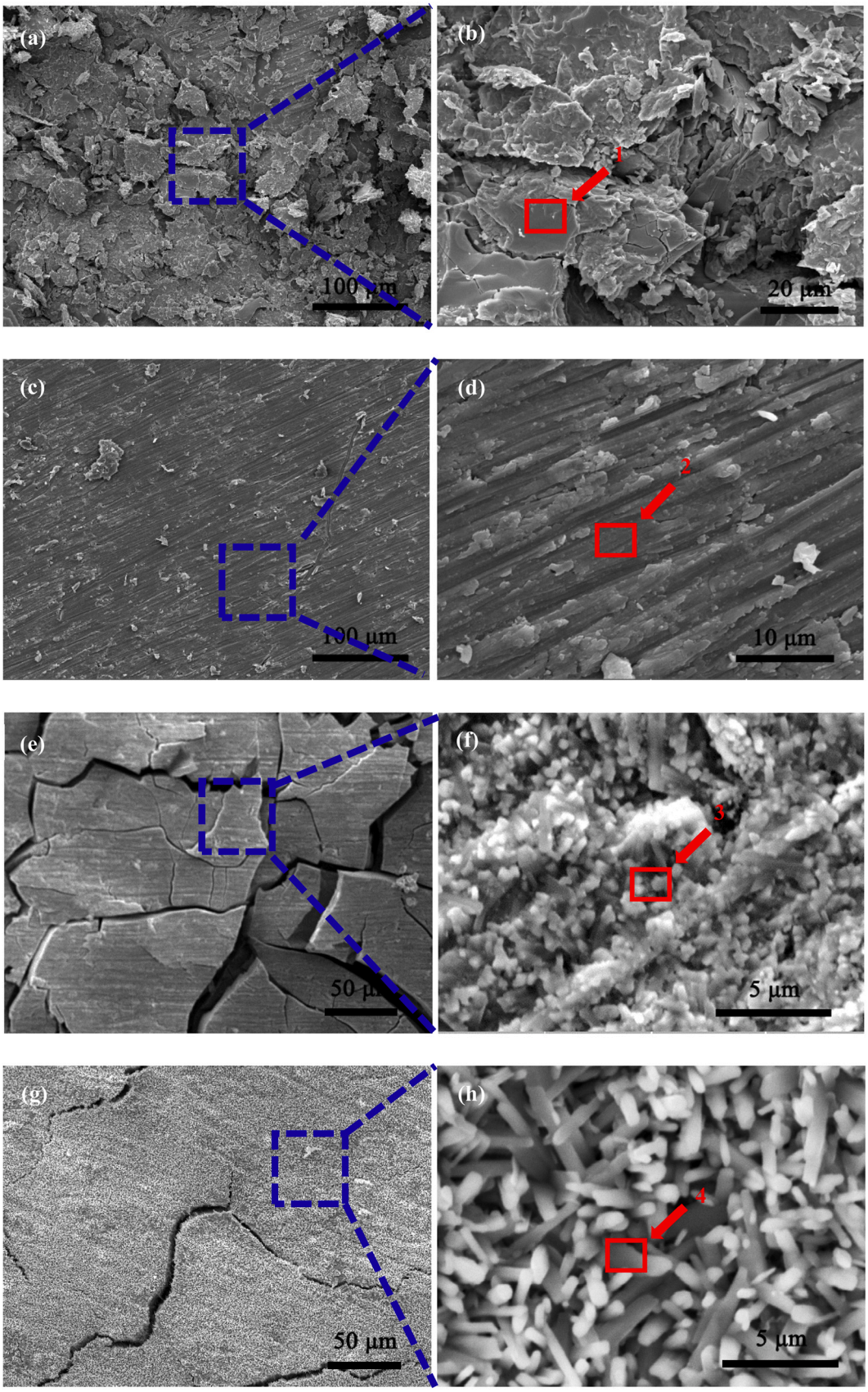


Fig. 5. Surface morphology of V3 and V11 alloys after isothermal oxidation at 500 °C for 200 h (a–d) and 800 °C for 24 h (e–h).

Table 4
The composition of microregion marked by numeric labels in Fig. 4.

Alloy	Oxidation process	site	Composition (at%)					
			Al	Ti	V	Zr	Nb	O
V3	500 °C@200 h	1	2.39	12.95	1.23	9.45	11.95	62.03
V11	500 °C@200 h	2	2.28	9.61	3.84	9.07	10.98	64.22
V3	800 °C@24 h	3	1.55	9.00	0.77	6.37	10.9	71.41
V11	800 °C@24 h	4	1.33	6.02	3.33	0.82	22.96	65.54

between V_2O_5 and Nb_2O_5 that was beneficial to the formation of Nb_9VO_{25}. Furthermore, the depletion of Nb in the interface of Nb oxide and matrix would further promote the external diffusion of Nb. Hence, when more V was contained in the sample, more Nb_9VO_{25} might be produced in the outer oxide layer; that is to say, more Nb could be detected in the outmost oxide layer of alloy with higher V content (e.g., V11 alloy). Combining the surface morphology variation with the increase of V, it is reasonable to believe that the columnar crystal in the oxide layer was Nb_9VO_{25} after oxidation at 800 °C. Furthermore, the low content of Zr in outmost layer of oxide layer reflects that there was a competition between V_2O_5 and ZrO_2 in the reaction with Nb_2O_5, and the formation of $Nb_2Zr_6O_{17}$ was restrained as more Nb_9VO_{25} was produced in the oxide layer.

Fig. 6 shows the cross sections of samples treated at different oxidation conditions. After exposed at 500 °C for 200 h, only one single layer of oxide was obtained on the surface of sample. For V3 alloy (Fig. 6a), the thickness of oxide layer was about 70 μm, serious peeling–off took place around the outmost oxide skin (Fig. S1), and the surface crack extended to the half thickness of the oxide layer. When the concentration of V was increased to 11 at%, the oxidation degree of alloy was alleviated at some extent. The corresponding average thickness and crack propagation depth of oxide layer were decreased to 30 and 5 μm, respectively (Fig. 6b). As the oxidation temperature was raised to 800 °C, the thickness of oxide layer was sharply increased to about 300–500 μm after oxidation for 24 h, and the oxide film consisted of two layers, i.e., the outer black layer and the internal gray layer. To V3 alloy (Fig. 5c), crisscrossing cracks spread over the whole oxide layer, and severe warped phenomenon took place in the oxide film. Furthermore, the cracks surrounded with dense tiny pinholes penetrated the fresh matrix along the interdendritic region as shown in Fig. S2. With regard to V11 ally, most of cracks initiated and propagated in the outer oxide layer at this stage, and few cracks could reach the upper part of internal oxide layer. Evidently, the oxide layer formed on the V11 alloy was tougher than that formed on the V3 alloy no matter what the oxidation temperature was, which suggests that the antioxidant performance of V11 alloy was superior to that of V3 alloy.

Fig. 7 shows the distribution of element in the cross section of oxide layer obtained at different oxidation conditions. Similar to previous researches about RCCA [33,42], the distribution of element in each oxide layer seemed to be uniform regardless of the oxidation temperature and the type of alloy, which suggests the oxide layers of V3 and V11 alloys were both oxide mixture. This phenomenon was further confirmed by TEM–EDS analysis toward the oxide layer obtained at 500 and 800 °C, as shown in Fig. S3 and S4. SEM–EDS results toward the cross section of oxide layer are presented in Table 5, including two types of chemical component for each testing location, one was conventional testing value, the other was estimated value calculated from the former one without O content. Ignoring the concentration of O, the atomic percent of metallic element in the oxide layer obtained at 500 °C was comparable to that in the matrix, which confirms that the oxidation procedure of NbTiZrAlV RCCA under this condition was controlled by the internal diffusion of O. As exposed at 800 °C, the estimated concentration of V in the oxide layer of V3 alloy was about 1.5–2.2 times of that in the matrix. However, the estimated value of V in the outer oxide layer of V11 alloy was only half of that in the internal oxide layer, whereas the estimated concentration of metallic elements in the internal scale

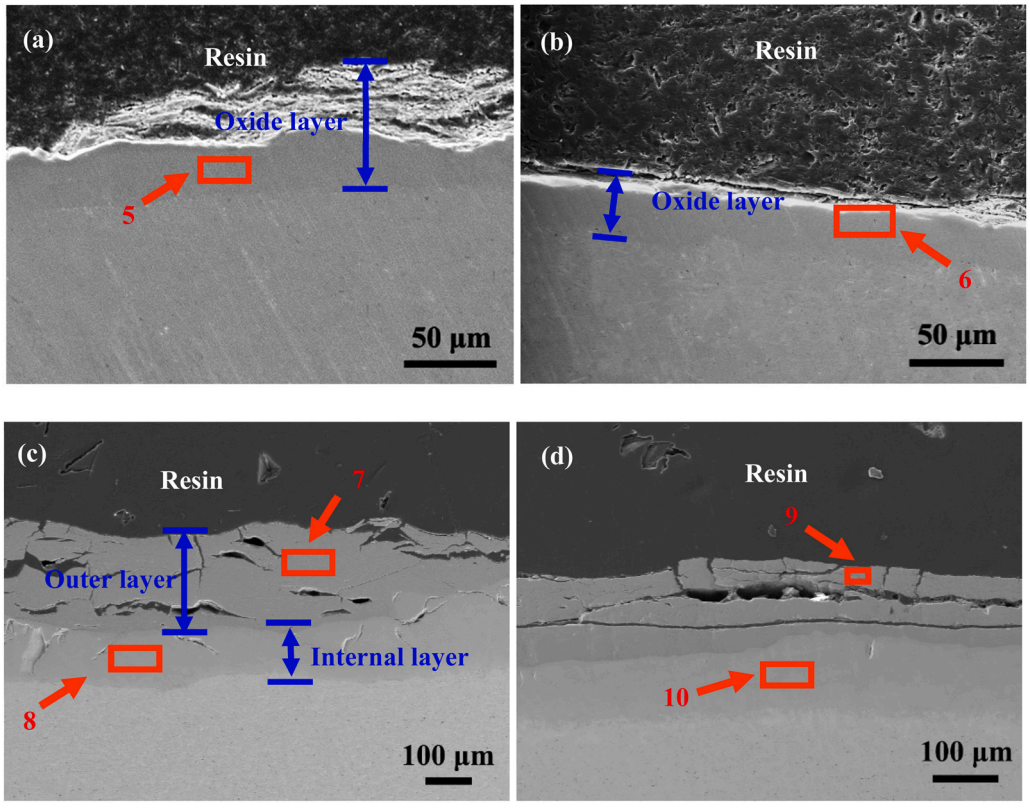


Fig. 6. Cross–sections of samples oxidized at different conditions: (a) V3, 500 °C for 200 h; (b) V11, 500 °C for 200 h; (c) V3, 800 °C for 24 h; (d) V11, 800 °C for 24 h.

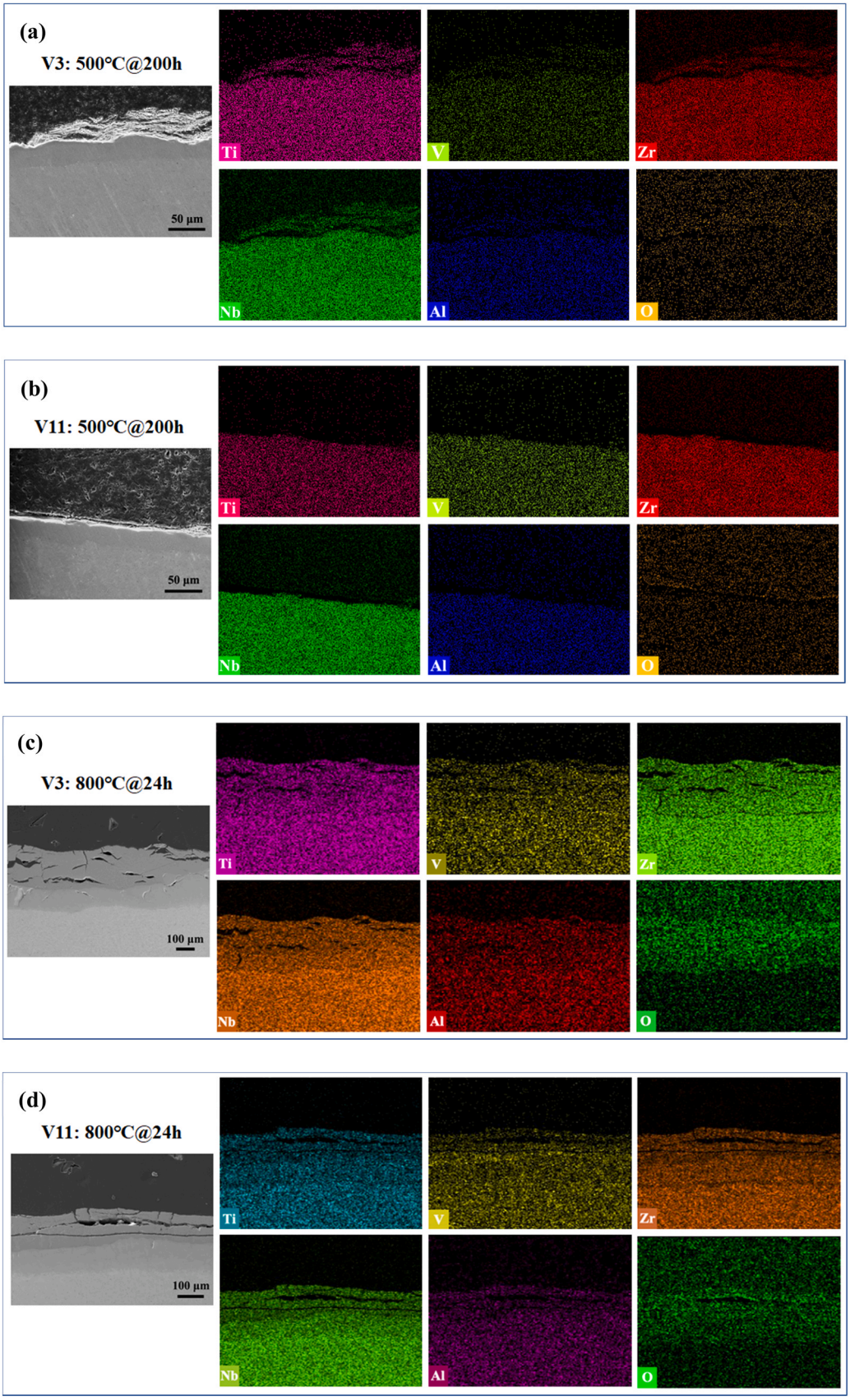


Fig. 7. Element distribution in the scale of samples after oxidation at different conditions: (a) V3, 500 °C for 200 h; (b) V11, 500 °C for 200 h; (c) V3, 800 °C for 24 h; (d) V11, 800 °C for 24 h.

Table 5
The chemical composition of microregion marked by numeric labels in Fig. 5.

Alloy	Oxidation process	site	Composition (at%)						Remark
			Al	Ti	V	Zr	Nb	O	
V3	500 °C@200 h	5	3.09	15.82	1.52	12.89	15.22	51.46	¤
V3	500 °C@200 h	5	6.37	32.59	3.13	26.56	31.36	–	*
V11	500 °C@200 h	6	4.93	19.77	7.66	20.4	23.13	24.43	¤
V11	500 °C@200 h	6	6.52	25.76	10.13	26.99	30.60	–	*
V3	800 °C@24 h	7	2.61	12.76	2.25	10.52	13.82	58.03	¤
V3	800 °C@24 h	7	6.22	30.41	5.36	25.07	32.94	–	*
V3	800 °C@24 h	8	3.34	15.94	3.82	14.7	17.72	44.48	¤
V3	800 °C@24 h	8	6.02	28.71	6.88	26.48	31.92	–	*
V11	800 °C@24 h	9	2.45	10.71	2.49	11.87	13.34	59.04	¤
V11	800 °C@24 h	9	6.00	26.21	6.09	29.05	32.65	–	*
V11	800 °C@24 h	10	3.20	14.13	6.08	14.29	18.49	43.83	¤
V11	800 °C@24 h	10	5.69	25.15	10.82	25.43	32.91	–	*

¤ Conventional testing value.
*Estimated value calculated from conventional testing value without O.

was approximate to the design value of V11 alloy. It should be noted that the change of V content in the oxide layer was inevitable to induce the concentration fluctuation of other metallic elements. The variation of chemical component for the oxide layer manifests the different oxidation behavior of two alloys at elevated temperature. And the corresponding oxidation mechanism would be discussed in details in the following section.

3.3. Oxidation mechanisms of NbTiZrAlV RCCA

The oxidation of alloy is essentially one process of atomic diffusion, including internal O diffusion and external componential elements diffusion. In our RCCAs, the diffusion kinetics of atom is greatly influenced by the mixing enthalpy and configuration entropy, which are speculated to be the key factors for the stabilization of solid solution. And minimizing mixing enthalpy and maximizing configurational entropy can be used to enhance the effect of sluggish diffusion in RCCA, which conductive to obtain desirable thermostability [43–45]. Based on the researches of Takeuchi [46] and Ye [47], mixing enthalpy and configuration entropy of V3 alloy were calculated to be $-4.76\ kJ\ mol^{-1}$ and 11.26 J/K·mol, whereas those values of V11 alloy were estimated to be $-5.28\ kJ\ mol^{-1}$ and 12.26 J/K·mol. Consequently, it can deduce that energy barrier for atom diffusion in V11 alloy was greater than that in V3 alloy, which endowed superior antioxidant properties to V11 alloy relative to the V3 alloy.

When the NbTiZrAlV RCCA was exposed at 500 °C, the oxidation behavior was mainly determined by the inward diffusion of O. Due to the high concentration of metallic elements and stability of oxides, TiO_2, ZrO_2 and Nb_2O_5 formed at the early stage of oxidation, and then ZrO_2 reacted with TiO_2 and Nb_2O_5 to produce Ti_2ZrO_6 and $Nb_2Zr_6O_{17}$, respectively. Meanwhile, the segregation degree of V in the interdendritic region of alloy was increased with the increase of V content that exacerbated the lattice distortion, leading to more significant atomic traps and blocks and thus higher diffusion activation energy of atom in the matrix of V11 alloy. Furthermore, the production rate of TiO_2 was reduced with the decrease of Ti concentration that slowed down the formation rate of Ti_2ZrO_6, which restrained the inward diffusion of O during the oxidation to some extent. Therefore, the oxide layer obtained at this condition mainly consisted of Ti_2ZrO_6 and $Nb_2Zr_6O_{17}$, and the thermostability of V11 alloy with higher V content was better than that of V3 alloy.

As oxidation temperature was increased to 800 °C, NbTiZrAlV RCCA exhibited different oxidation behavior deriving from the formation and volatilization of V_2O_5. Firstly, the volatilization of V_2O_5 increased the local partial pressure around the interdendritic region that greatly accelerated the reaction between Nb_2O_5 and V_2O_5 to form Nb_9VO_{25}. Secondly, the volatilization of V_2O_5 gave rise to the generation of pinholes in the interdendritic regions, which promoted the internal O diffusion as well as the oxidation of Nb and Zr adjacent to these regions, and eventually led to the formation of $Nb_2Zr_6O_{17}$. Thirdly, the pinholes in the matrix would act as the source for crack initiation and brought about the crack propagation along the interdendritic region, inducing further growth of oxide layer and degradation of protection of oxide layer for matrix. For V3 alloy, the estimated content of V in the oxide layer was about two time of that in the matrix, demonstrating a certain amount of V in the oxide layer was attained by the outward migration of V from the adjacent matrix. And the external V diffusion would leave high-concentration vacancy layer between matrix and oxide layer, and cause the variation of lattice parameter in the microregion, leading to the formation of internal stress in the front of matrix. Thus, cracks

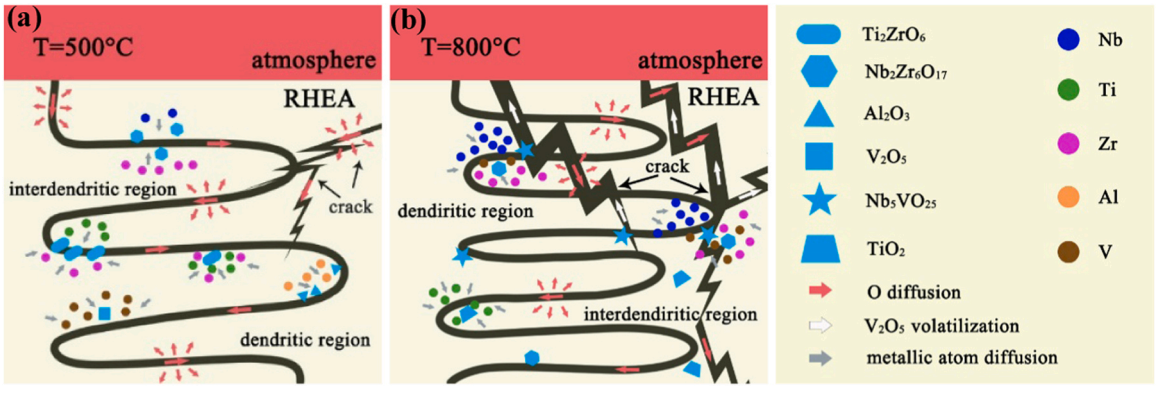


Fig. 8. Schematic illustration of the oxidation behaviors of V3 and V11 alloys at 500 (a) and 800 °C (b).

would quickly initiate in the V–depletion layer and continuously penetrated the fresh matrix via the synthetic action of internal and thermal stress during the oxidation process, and in turn accelerated the oxidation rate of material and aggravated the peeling–off of oxide layer. Therefore, dense cracks were observed in oxide layer as well as in the substrate. To V11 alloy, the estimated content of V in the outer scale was lower than that of substrate, whereas that estimated value in the internal oxide layer was comparable to that of matrix. This phenomenon suggests that external V migration merely took place in the outmost layer of matrix, and the external V diffusion rate in the V11 alloy was distinctly slower than that in the V3 alloy. As a result, the formation of pinhole and vacancy induced by the volatilization of V_2O_5 were greatly reduced in the V11 alloy, and only took place in the outer scale. Additionally, the internal oxide layer could act as one stress buffer layer, which coordinated the thermal deformation difference between outer oxide layer and matrix. That is, the V11 alloy was equipped with enhanced crack resistance, and the vast majority of cracks only initiated and propagated in the oxide layer. As mentioned above, it can be concluded that the NbTiZrAlV RCCA with higher V content possessed superior thermal stability, and the oxidation behavior of V3 and V11 alloys during exposure at 500 and 800 °C is shown schematically in Fig. 8.

4. Conclusion

In this work, two kinds of NbTiZrAlV RCCA with different V concentration were fabricated and corresponding isothermal oxidation behavior was investigated in detail. The antioxidant performance of NbTiZrAlV RCCA was obviously improved by the increase of V concentration due to the enhanced effect of sluggish diffusion, deriving from the reduction of mixing enthalpy and increase of configurational entropy. When NbTiZrAlV RCCA was oxidized at 500 °C, the oxidation process of alloy was determined by the internal O diffusion. And single oxide layer mainly consisted of Ti_2ZrO_6 and $Nb_2Zr_6O_{17}$ that formed after thermal exposure for 200 h. The mass gains of alloys with the V content of 3 and 11 at% were 0.042 and 0.018 mg/mm^2, and the average thickness of oxide layer was 70 and 30 µm, respectively. When NbTiZrAlV RCCA was exposed at 800 °C, the oxidation behavior of NbTiZrAlV RCCA was controlled by the combination action of inward O diffusion and outward V diffusion. And complex oxide layers composed of $Nb_2Zr_6O_{17}$, Nb_9VO_{25} and TiO_2 were obtained after oxidation for 24 h. As the content of V increased from 3 to 11 at%, the mass gain of oxidized alloy decreased from 1.135 to 0.514 mg/mm^2, and the average thickness of oxide layer was reduce by two–fifths.

CRediT authorship contribution statement

Yi Lin: Project administration, Investigation, Writing – review & editing. **Yi Guo:** Investigation. **Quan Dong:** Investigation. **Rui Huang:** Investigation. **Jun Tan:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jallcom.2022.164180.

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